School of Electrical Engineering and Computer Sciences
National University of Sciences and Technology

DEPARTMENT OF Computing

Industry Advisory Board Meeting Minutes

Held on May 24, 2023

In attendance

S.No.	Name	Designation/Organization/Company
1.	Dr. Muhammad Moazam Fraz (In-Chair)	Associate Professor, HoD AI and Data Science, Department of Computing, NUST-SEECS
2.	Dr. Mehdi Hussain	Assistant Professor, NUST-SEECS
3.	Dr. Muhaamad Imran Malik	Associate Professor, Head CS Department, NUST-SEECS
4.	Dr. Rafia Mumtaz	Professor, Head SE Department, NUST-SEECS
5.	Dr. Shah Khalid	Assistant Professor, FYP Coordinator CS, NUST-SEECS
6.	Dr. Hashir Moheed	Assistant Professor, FYP Coordinator SE, NUST-SEECS
7.	Dr. Daud Abdullah Asif	Assistant Professor, NUST-SEECS
8.	Dr. Ahsan Saadat	Assistant Professor, NUST-SEECS
9.	Dr. Yasir Faheem	Associate Professor, NUST-SEECS
10.	Dr. Muhammad Bilal	Assistant Professor, UG-Coordinator CS, NUST-SEECS
11.	Dr. Zuhair Zafar	Assistant Professor, NUST-SEECS
12.	Mr. Muhammad Tashfeen	Industrial Liaison officer, NUST-SEECS
13.	Mr. Umer Nasim	Software Engineer/ Department Coordinator, NUST-SEECS
14.	Dr. Sohail Sarwar	President SEECS Alumni Association
15.	Mr. Abdullah Fatik	Director Technology, StartifyFun
16.	Mr. Shahryar Khan	CEO, Swam Technologies
17.	Dr. Shujaat Hussain	Assistant Professor, FAST NUCES
18.	Dr. Sher Serwar	PD Public Sector / AP NAMAL
19.	Ms. Samina Ikram	VP, Crescent School
20.	Mr. Muhammad Arsalan	COO, Bish Technology
21.	Mr. Asad Asif	Director Operations, ECI (Pvt) LTD
22.	Ms. Sarah Shafaqat	Director Marketing, Dream Soft Tech
23.	Mr. Fahad Akbar	Team Lead, Tech Insects

24.	Mr. Malik Hamza	HoD Info Security, Mobilink
25.	Dr. Ali Irtaza	Assistant Professor, IST University
26.	Mr. M. Fawad Bary	Associate Director, S&P Global
27.	Mr. Umair Ali	Associate Director, S&P Globl
28.	Mr. Nauman Faridi	SVP Operations, Walee Technologies
29.	Mr. Subhan Khan	CEO, Big4Technology
30.	Ms. Sundas Khurshid	Scrum Master, S&P Global
31.	Mr. Imran Ahmad	Manager Internet Governance, Ministry of IT & Telecom
32.	Mr. Tayyab Hushmi	Deputy Director, Standards & Policy, NITB Ministry of IT & Telecom
33.	Mr Shamyl Bin Mansoor	Chief Scientific Officer, LearnOBots

Venue: Board Room CIPS

Meeting time: 1400 hrs – 1600 hrs

Agenda Points / Discussion

- The Industry Advisory Board (IAB) meeting comprises industry representatives from prestigious companies and faculty members from SEECS. The IAB meetings aim to enhance the academia-industry linkage and its impact through industrial feedback on the design, development, delivery, and evolution of the curriculum of computing degree programs.
- Dr. Muhammad Moazam Fraz briefed the house about the department, the degree programs, the vision and mission of the department and how the curriculum is designed to achieve this vision and mission.
- The industry representatives briefed the house about their companies and the different products and technologies on which they are working.
- The meeting started with a brief introduction of the board members, followed by a discussion on the following points.
  - Launching of the new degree program, BS in Artificial Intelligence
  - Increase in annual intake of BSCS program from Fall 2023
  - Enlighten the Industry about Outcome Based Education (OBE) and how the undergraduate degree programs (BESE, BSCS, BSDS and new program BSAI) are designed to align with the OBE paradigm, such that better equipped and skilled graduates are produced who can meet industry demands.
  - Revised Program Educational Objectives (PEOs) of BESE, and PEOs of BSCS, BS DS and BSAI degree programs
  - Program Learning Outcomes (PLOs) of BESE, BSCS, BSDS and BSAI
  - How the curriculum of these degree programs is enhanced based on suggestions from previous IAB meetings and the input taken from the industry on various other occasions
- Dr Ahsan gave a brief introduction about SEECS, the BESE curriculum and about different facilities that are offered to students. During the presentation, he presented the revised BESE PEO statements, and the KPIs to measure the attainment of PEOs. The detailed documentation of the revision process (revision of PEO statements, revision of KPIs for PEOs and revision of survey forms) was provided to the participants in hard copy for endorsement. The same documents are attached as Annexures. The IAB participants recommended the revised PEOs, KPIs and survey forms.

He also presented the steps that are being taken for the improvement of the quality of the FYDPs, continuous quality improvement and curriculum modernization. He further asked the house to give suggestions on the different topics that can help the student's learning process.

- Dr Moazam opened the house for the discussion.
- Dr Shujaat Hussain praised the arrangements for the open house. He further added that this time students focused more on IoT-based projects. AI-related projects need more improvements. For course offerings, he said that it seems aligned with HEC/PEC guidelines. He suggested that faculty placement in the industry is a good idea for making faculty aware of related latest technologies.
- Ms Sundas Khurshid added that the soft skills of students should be improved, and different types of training should be conducted for them.
- Dr Sohail suggested that more advanced level courses should be introduced in the curriculum. He added that recently 250 plus students got benefits from CISCO network academy, so these types of collaborations should be expedited. Students should focus more on the vertical depth of the projects and the commercialization aspect that was missing in the projects. He further added that students should frequently use GitHub and LinkedIn.
- Mr Fawad suggested that students should focus more on enterprise-level skills and projects should be more focused on product-based solutions and students should work on industrial projects. Dr Moazam briefed him that we are already focusing on multidisciplinary projects. Dr Imran Malik added that many students worked on industrial projects as their FYDPs.
- Mr Tayyab Hussain suggested that industry representatives be invited as guest speakers to give hands-on experience on a specific topic.
- Mr Nauman Faridi pointed out that all the stakeholders should work to minimize the industry-academia gap. He suggested that a mechanism should be developed where we can produce graduates who have problem-solving and learning abilities skills. Students should be taught more about ethical and professional practices. He added that the institute should focus on industry talks in the classes.
- Dr Moazam briefed the house about the launch of new degree programs BS in Artificial Intelligence and BS in Data Science. In general, the launch of the program is highly appreciated by the IAB members. Some specific comments from the industry are as under.
    - Dr Sohail Asghar appreciated the initiative and said that the introduction of the BS Artificial Intelligence Program at SEECS comes at a crucial time when there is a growing demand for specialized AI-skilled graduates with strong problem-solving and software development abilities. In today's rapidly advancing technological landscape, artificial intelligence has become a key driver of innovation across various industries. By offering this program, SEECS will be able

to address the need for professionals who possess a deep understanding of AI concepts and techniques, coupled with the practical skills to develop robust software solutions. This program will equip students with the necessary knowledge and expertise to tackle complex challenges in the field of AI, empowering them to make significant contributions to the ever-evolving landscape of artificial intelligence. The launch of this program is a timely initiative that will play a vital role in meeting the industry's demand for skilled AI professionals and shaping the future of AI innovation.

- Mr Shamyl Bin Mansoor wholeheartedly endorsed the launch of the BS Artificial Intelligence Program at SEECS, recognizing the pressing need for specialized graduates in the field of AI. As businesses and industries increasingly rely on data-driven technologies and automation, there is a growing demand for professionals who possess the skills and expertise to optimize processes and drive innovation through AI. This program will equip students with a deep understanding of artificial intelligence, machine learning, and data analytics, enabling them to harness the power of AI to solve complex problems, make informed decisions, and enhance business operations. By producing a new generation of AI-skilled graduates, SEECS will contribute to the growth and competitiveness of industries, fostering economic development and technological advancements in the region.
- Mr Nauman Fareedi stated that it's good to know about the launch of the BS Data Science Degree Program at SEECS, recognizing the pressing need for skilled graduates in the field of Data Science within the local industry. With the exponential growth of data-driven technologies and the increasing reliance on data analysis for decision-making, there is a high demand for professionals equipped with specialized knowledge in Data Science. This program will bridge the gap between academia and industry by providing students with a comprehensive understanding of data management, statistical analysis, machine learning, and data visualization. By offering a curriculum tailored to the evolving needs of the industry, the BS Data Science Degree Program at SEECS will produce graduates who possess the practical skills and theoretical foundation necessary to address complex real-world challenges. This initiative will undoubtedly contribute to the development of a highly skilled workforce, fostering innovation and driving economic growth in the region.

● Dr Moazam appreciated the feedback from the industry and mentioned that due efforts would be taken by the department to implement the suggested points.

DoC Department Coordinator
May 30, 2023

Mr Taufique ur Rehman
DLQEC SEECS
May 30, 2023

Annex-A

Endorsement Form for Continuous Quality Improvement - PEOs

(NUST-SEECS)

First Name:	
Last Name: Contact Information (Phone No): Email:	
Name of the Organization/Company:	
Designation:	
City & Country:	

☐ **I endorse the revised PEO statements, KPIs for PEOs and the survey forms prepared by Software Engineering Department at SEECS, NUST.**

Comments (if any):

☐ **I do not endorse the revised PEO statements, KPIs for PEOs and the survey forms prepared by Software Engineering Department at SEECS, NUST.**

Comments (if any):

Signature_____

Endorsement Form for Continuous Quality Improvement - PEOs

(NUST-SEECS)

First Name:	
Last Name: Contact Information (Phone No): Email:	
Name of the Organization/Company:	
Designation:	
City & Country:	

☐ **I endorse the revised PEO statements, KPIs for PEOs and the survey forms prepared by Software Engineering Department at SEECS, NUST.**

Comments (if any):

☐ **I do not endorse the revised PEO statements, KPIs for PEOs and the survey forms prepared by Software Engineering Department at SEECS, NUST.**

Comments (if any):

Signature_____

Annex-B

NUST
NATIONAL UNIVERSITY
OF SCIENCES & TECHNOLOGY

SCHOOL OF ELECTRICAL ENGINEERING &
COMPUTER SCIENCE (SEECS)

Industry Advisory Board Meeting

24 May 2023

1

AGENDA

SEECS Introduction

Washington Accord – PEC Visit

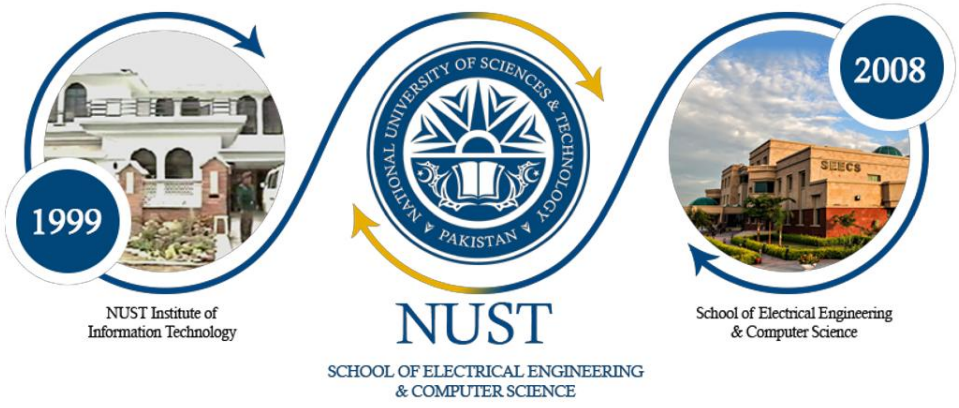
PEO, Alumni and Employer Survey Revision and Endorsement

Curriculum Improvements and Efforts to Meet Industry Needs

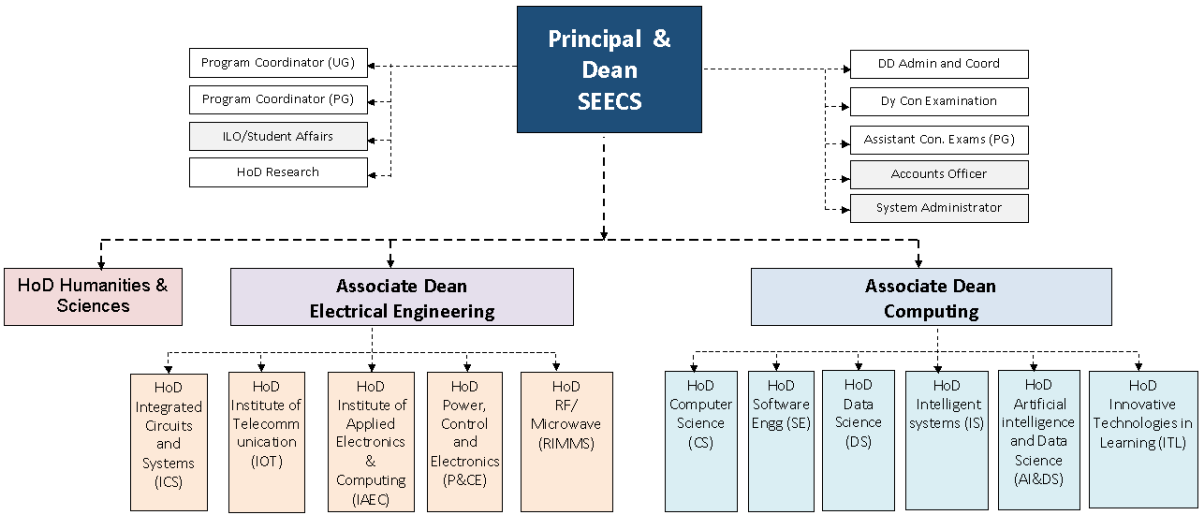
Specific Improvements Since Open-house 2022

SEECS - WHERE CYBER AND PHYSICAL WORLDS MEET

SEECS - HISTORY



SEECS - ORGANIZATION



SEECS – AT A GLANCE



SEECS – ACADEMIC PROGRAMS

UG Programs (03)	MS Programs (06)	PhD Programs (04)
Electrical Engineering (BEE)	Electrical Engineering (MSEE)	Electrical Engineering (PhDEE)
Computer Science (BSCS)	Computer Science (MSCS)	Computer Science (PhDCS)
Software Engineering (BESE)	Information Technology (MSIT)	Artificial Intelligence (PhDAI)
BS Data Science (BS DS)	Innovative Technologies in Learning (MSITL)	Information Technology (PhDIT)
	Information Security (MSIS)	Information Security (PhDIS)
	Data Science (MSDS)	
UG Students: 1594	**MS Students: 951**	**PhD Students: 101**

WASHINGTON ACCORD

Pakistan Engineering Council (PEC) became the full signatory of WA in 2017

To maintain the quality of accreditation system in each signatory jurisdiction, WA Periodic Review of full signatories is held after every five years

WA team will visit NUST in October 2023

PEC has selected three NUST programs for review

BE Software Engineering program at SEECS

PROGRAM EDUCATIONAL OBJECTIVES

Revision of PEOs, employer survey and alumni survey

Within five years of graduation, the graduates of SEECS Bachelor of Software Engineering should:

PEO-1: Demonstrate sound software engineering knowledge and skill set.

PEO-2: Lead diverse teams and work collaboratively to develop innovative solutions.

PEO-3: Conduct professional practice for sustainable development with due consideration for societal, ethical and environmental aspects.

CURRICULUM AND LEARNING PROCESS 2022-23

Periodic CQI meetings to improve the curriculum

Revision of the CLOs and BT levels of several courses

Additional CLOs to cover psychomotor and affective domains to balance overall PLO mapping

Revision of lab rubrics to assess psychomotor and affective domains

Creation of a standardized lab rubric for a one-on-one mapping of the rubric items with CLOs

Formal training sessions for LEs and FMs

CURRICULUM MODERNIZATION

Inclusion and Revision of Courses to Meet Industry Needs

Occupational Health and Safety	Network Security
Computer Architecture and Logic Design	Artificial Intelligence
Cloud Computing	Machine Learning
Human Computer Interaction	Advanced DBMS
Information Security	Advanced Programming
Design and Analysis of Algorithms	Computer Networks

11

CURRICULUM MODERNIZATION - ELECTIVES

Software Eng- Core Depth Elective Courses		
Sr.	Title	CHs
1	Big Data Analytics	2+1
2	Deep Learning	3+1
3	Agent Based Software Engineering	2+1
4	Visual Programming	2+1
5	Real Time Systems	2+1
6	Data Encryption and Security	2+1

CS – Elective Courses		
Sr.	Title	CHs
1	Web Engineering	3+1
2	Deep Learning	3+1
3	Network Security	3+1
4	Computer Vision	3+0
5	Natural Language Processing	3+0
6	Mobile Application Development	3+1
7	Image Processing	3+1

Multi Disciplinary Engineering Electives		
Sr	Title	CHs
1	Internet of Things	3+0
2	Embedded Systems	2+1

12

SUSTAINABLE DEVELOPMENT



PROJECT BASED LEARNING

FYDP Rubric Revision

FYDP mapping to SDGs

FYDPs in collaborating with industry to solve real world problems

Complex engineering problems and trainings

Open-ended labs

Addition of lab components in courses – updating the % contribution of practical implementation

INFRASTRUCTURE GROWTH

Construction of a new academic block has been planned in 2023

- 30,000 sq. ft. space for class rooms, labs, faculty/ support offices

Establishment of a one general purpose lab and one additional FYDP lab to facilitate practical learning of students

2 x new labs in collaboration with industry

- Walee-NUST centre of excellence
- OneScreen research lab
- Industrial partners have also provided more than PKR 5M value equipment to SEECS

UPSKILL / RESKILL TRAININGS AND BOOTCAMPS

Trainings and Workshops are held quite frequently to keep the students apprised about latest industrial trends

- AI and Data Science
- Business Data Analytics
- Machine / Deep Learning
- Mobile App Development
- Internet of Things (IoT)
- Cloud Computing
- Parallel and Distributed Computing
- Network security
- BlockChain
- Computer Vision / NLP

16

SOFT SKILLS

Oral presentation is now mandatory in all courses to improve communication skills of students

Mock interviews are arranged for final year students to improve their interviewing skills

17

DEVOPS CULTURE

The students are encouraged to use software version control tools, automated build tools and IDEs in their projects and assignments

The students are offered a course on Open Source Systems

18

PARTICIPATION IN COMPETITIONS

Students are encouraged to participate in national and international competitions like Microsoft Imagine Cup, Huawei ICT Innovation Competition, Google's coding competitions, Cisco Global Problem Solver Challenge etc

Recently our FYDP have won first position and a merit award in the categories of IoT & Digital Systems, and Software Applications at 20th Edition of COMPEC 2023

19

WALEE-NUST CENTRE OF EXCELLENCE

SE department has initiated collaboration with Walee Technologies

For this purpose, a joint lab in SEECS has been established

This lab offers opportunities such as paid internships, training, and scholarships to students

In addition, faculty is financially compensated for supervising the company projects with the help of students

20

IMPROVEMENTS SINCE LAST OPEN HOUSE (JUNE, 2022)

Industry suggested to introduce DevOps cycle to students

Action: Students are introduced to DevOps cycle during software design courses

For the industry talks new trends in industry should be introduced to students

Action: Several seminars have been conducted by IEEE-SEECS society on emerging topics such as Quantum computing, Industrial IoT

Industry suggested that product-based projects should be encouraged

Action: Students are encouraged to take projects in collaboration with industry. Several product-based projects are currently being undertaken by final year students.

Industry suggested that FYDPs should be encouraged to file patents

Action: Students are encouraged to file patents for their FYDPs. NUST IP office provides full support in this regard. 21

IMPROVEMENTS SINCE LAST OPEN HOUSE (JUNE, 2022)

Industry suggested that students should be given training on industry relevant technologies

Action: Industrial trainings are regularly offered to students to enhance their technical skills.

Industry suggested to add IoT related course

Action: Course on IoT is now a part of the curriculum. In addition to provide a lab component in this course, the working paper is in progress.

Industry suggested to improve student's entrepreneurship skills

Action: Students are encouraged to take internships at different start-ups in NSTP. Talks by experts are arranged that specifically target entrepreneurship skills.

Industry suggested engagement in more interdisciplinary projects

Action: Several funded projects undertaken by faculty are interdisciplinary. For example, "Building Capacity to Improve Air Quality in South Asia" and "IoT and AI Enabled System for Crop Disease Monitoring and Prediction. 22

CONCLUSION - ACTIONS

Suggest topics / content for addition in curriculum especially for programming and industry-oriented courses

Deliver guest lectures for selected courses

Offer internships and trainings for students

Short-term placements for faculty

Engage students in industry projects by offering FYDPs

23

Annex-C

Alumni Survey

(NUST-SEECS)

The purpose of this survey is to obtain alumni input on the quality of education they received at NUST. We seek your help in completing this survey**.**

A: Excellent (5) B: Very Good (4) C: Good (3) D: Fair (2) E: Poor (1)

		A (5)	B (4)	C (3)	D (2)	E (1)
1	knowledge compared to your peers and colleagues					
2	Competence and skillfulness compared to your peers and colleagues					
3	Involvement in research and innovative activities					
4	Independent project handling skills					
5	Participation in professional development related activities					

		Yes	No
6	Employment within one year of your graduation		
7	Pursuing higher education		

8	Incorporation of UN SDGs in your projects / services / solutions / products, etc.		
9	Involvement in projects addressing societal and/or environmental issues		

Alumni Information

1. Name:	
2. Year of Graduation	
3. Name of Current Organization	
4. Position in Organization	
5. Served Organizations/Private firms	
5. Email	
6. Telephone	

Annex-D

Employer Survey

(NUST SEECS)

Dear Employer of NUST graduates,

In order to continue to serve your professional requirements, it is imperative that the BESE program offered at our institution is up to national and international standards. For this purpose, your feedback is of prime importance for us to make continuous quality improvements. The purpose of this survey is to obtain your input on the quality of education NUST is providing and to assess our academic programs from the point of view of your needs and demands. We seek your help in completing this survey.

Please rank the following traits of your employee who has graduated from NUST with a BESE degree. Kindly provide a fair evaluation.

A: Excellent (5) **B: Very Good (4)** **C: Good (3)** **D: Fair (2)** **E: Poor (1)**

		A (5)	B (4)	C (3)	D (2)	E (1)
1	Technical knowledge					
2	Competence and skillfulness					
3	Communication and presentation skills					
4	Ability to work in team environment					
5	Independent project handling skills					

6	Awareness of ethical values in professional life					
7	Discharge of professional obligations					

Employers Contact Information

1. Company/Organization	
2. First Name	
3. Last Name	
4. Position/Appointment	
5. City & Country	
6. Email	
7. Telephone	